An Incremental Bangladesh Socio Economics Data Integration Model

Database Systems Project

Scrum Level-01
Scrum Group-14

Report submitted August, 2026

A project submitted to Prof. Dr. Rudra Pratap Deb Nath, Department of Computer Science and Engineering, Chittagong University (CU) in partial fulfillment of the requirements for the Database Systems Lab course. The project is not submitted to any other organization simultaneously.

Table 1: Details of Group-01

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Preface

This report presents the work carried out by Group 14 (UNKNOWN) as part of the Database System (CSE 413) course offered by the Department of Computer Science and Engineering, University of Chittagong. The project focuses on the Socio-Economic Sector of Bangladesh, with particular emphasis on collecting, organizing, modeling, and integrating socio-economic data into a unified database design. The primary objective of this project is to collect, organize, model, and integrate socio-economic data from various reliable sources to develop a comprehensive representation of the socio-economic condition of Bangladesh.

The project involves extensive research, data acquisition, dataset development, Entity-Relationship (ER) modeling, and ER integration. To accomplish this objective, data are collected from multiple reliable sources, including the Bangladesh Bureau of Statistics (BBS), UNESCO, BBS HIES (Household Income and Expenditure Survey), and other relevant publicly available databases and websites. The collected information is organized into numerous Excel datasets. Individual ER diagrams are then developed for each dataset. Finally, the separate ER models are integrated into a comprehensive ER diagram representing the Socio-Economic Sector database of Bangladesh.

The journey of this project has been anything but straightforward. What you see in this report is the culmination of countless hours of working with diverse datasets, inconsistent data formats, complex statistical information, and the ever-present challenge of making sense of real-world socio-economic data. This report is not just a documentation of our technical work—it is also an account of the challenges, learning experiences, collaborative efforts, and achievements that shaped our project journey.

The project provides practical experience in database design, data integration, conceptual modeling, data analysis, teamwork, and project management. It also enhances our understanding of real-world database development processes and the challenges associated with handling large-scale, heterogeneous socio-economic datasets. Through this project, we gained valuable in-

sights into how socio-economic information can be systematically organized and represented within a database system.

We would like to express our sincere gratitude to our course instructor, Prof. Dr. Rudra Pratap Deb Nath, for his continuous guidance, valuable feedback, and support throughout the project.

Name of the member 1 : Shamim Hosen
Name of the member 2 : Naim Uddin Mohammad Forhan
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Abstract

The socio-economic condition of Bangladesh is reflected through a wide range of indicators related to population, income, poverty, employment, education, health, agriculture, living standards, and other dimensions of development. However, these data are distributed across different organizations, surveys, reports, and online databases, making it difficult for researchers, policymakers, and other stakeholders to access, integrate, and analyze socio-economic information from a unified platform. The primary motivation of this project is to develop an integrated database model that can organize and represent diverse socio-economic information of Bangladesh in a structured and meaningful way. The main problem addressed in this project is the fragmentation and heterogeneity of socio-economic data across different sources, where differences in data structures, formats, definitions, and levels of detail create difficulties in combining and analyzing the information. To address this gap, data are collected from reliable sources including the Bangladesh Bureau of Statistics (BBS), Bangladesh Bureau of Statistics Household Income and Expenditure Survey (BBS HIES), United Nations Development Programme (UNDP), UNESCO, World Bank, FAOSTAT, and other relevant sources. The collected data are organized into structured datasets, followed by data analysis, conceptual modeling, individual Entity-Relationship (ER) diagram development, and integration of the individual ER models into a comprehensive socio-economic database model. The proposed solution provides a unified conceptual representation of interconnected socio-economic information of Bangladesh, covering relevant domains such as population, households, income and expenditure, poverty, employment, education, health, agriculture, and other socio-economic indicators. The solution will be evaluated based on the completeness, consistency, correctness, and integration of the developed datasets and ER models, including the proper identification of common entities and relationships and the reduction of unnecessary redundancy. The significance of this project lies in demonstrating the practical application of database design, conceptual modeling, and data integration techniques for organizing complex socio-economic information. However, the project may be limited by differences in data definitions, formats, time periods, availability, and coverage among the selected sources. In future work, the database can be extended with more recent and detailed datasets, automated data updating, analytical and visualization tools, and decision-support capabilities for researchers, policymakers, and other stakeholders.

1 Introduction

This document describes the design and development of an integrated Socio-Economic Database Model for Bangladesh, undertaken as part of the Database System (CSE 413) course offered by the Department of Computer Science and Engineering, University of Chittagong. The project aims to apply relational database theory, conceptual modeling, data integration, Entity-Relationship (ER) modeling, and database design techniques learned throughout the course to a realistic socio-economic data management problem. Socio-economic information of Bangladesh is generated and maintained by various national and international organizations, including the Bangladesh Bureau of Statistics (BBS), Bangladesh Bureau of Statistics Household Income and Expenditure Survey (BBS HIES), United Nations Development Programme (UNDP), UNESCO, World Bank, FAOSTAT, and other relevant sources. Since these data are distributed across different sources and often follow different structures, formats, definitions, and time periods, this project focuses on collecting, organizing, modeling, and integrating the information into a unified conceptual database model.

1.1 Background and Motivation

Socio-economic information plays a vital role in understanding the overall development and living conditions of a country. In Bangladesh, important information related to population, households, income and expenditure, poverty, employment, education, health, agriculture, living standards, and other development indicators is collected and published by different organizations such as BBS, BBS HIES, UNDP, UNESCO, World Bank, and FAOSTAT. However, these data are often distributed across separate reports, surveys, websites, statistical databases, and datasets, and may use different formats, definitions, classifications, and levels of detail. This fragmentation makes it difficult for researchers, policymakers, development organizations, and other stakeholders to access and analyze socio-economic information from a unified structure. To address these challenges, this project proposes an integrated database model that collects socio-economic data from multiple reliable sources, organizes them into structured datasets, identifies common entities and relationships, and integrates the individual models into a comprehensive Socio-Economic Database Model for Bangladesh. The significance of the proposed solution lies in providing a structured conceptual representation of heterogeneous socio-economic information that can support data retrieval, comparison, analysis, research, and future data-driven decision-making.

1.2 Problem Statement

Precisely state your Socio-economic data of Bangladesh are maintained and published by numerous organizations, resulting in fragmented and heterogeneous information repositories. Data related to population, households, income, expenditure, poverty, employment, education, health, agriculture, and other socio-economic indicators are often distributed across different datasets and sources, making it difficult to establish relationships among them and perform integrated analysis. The problem is to design a relational database model capable of representing the major socio-economic entities and their relationships, including demographic entities, households, geographic locations, economic indicators, income and expenditure information, employment, education, health, agricultural information, and development indicators, while reducing redundancy and maintaining consistency among integrated datasets.

1.3 System Definition

A computerized system used to organize and represent Bangladesh's socio-economic information by integrating data from multiple reliable sources and maintaining relationships among demographic, household, economic, employment, education, health, agricultural, geographic, and development-related entities using conceptual relational modeling and integrity constraints to reduce data redundancy, prevent anomalies, and ensure consistency.

1.4 System Development Process

The system is developed using a Scrum-based Agile methodology through three sprints with a four-person team. The first sprint focuses on researching socio-economic information and collecting data from sources such as BBS, BBS HIES, UNDP, UNESCO, World Bank, FAOSTAT, and other relevant websites, reports, and datasets. The second sprint focuses on cleaning, organizing, transforming, and developing structured datasets from the collected information. The third sprint focuses on developing individual Entity-Relationship (ER) diagrams for the datasets, identifying common entities and relationships, and integrating the individual ER models into a comprehensive Socio-Economic Database Model. Detailed activities and outcomes of each sprint are presented in the subsequent sections of the report.

1.5 Organization

This report is organized into several sections to present the complete development process of the Socio-Economic Database Model. Chapter 1 introduces the project, its background and motivation, problem statement, system definition, development process, and organization of the report. Chapter 2 presents the project structure, team management, and responsibilities. Chapter 3 describes the activities related to research, data collection, dataset development, and initial ER modeling. Chapter 4 presents the data cleaning, transformation, and refinement activities. Chapter 5 discusses the development and integration of individual ER diagrams into the final Socio-Economic Database Model. The subsequent chapter(s) discuss deployment, maintenance, collaboration, learning outcomes, and other project-related activities. Finally, the conclusion summarizes the project outcomes, limitations, and possible directions for future work.7.

2 Project Structure and Management

2.1 Project Overview

This project is conducted as part of the Database System (CSE 413) course under the Department of Computer Science and Engineering, University of Chittagong. The project focuses on designing an integrated database model for the Socio-Economic Sector of Bangladesh. The project involves research, data collection, dataset development, data transformation, Entity-Relationship (ER) modeling, ER integration, validation, and documentation. Socio-economic data are collected from multiple reliable sources, including the Bangladesh Bureau of Statistics (BBS), Bangladesh Bureau of Statistics Household Income and Expenditure Survey (BBS HIES), United Nations Development Programme (UNDP), UNESCO, World Bank, FAOSTAT, and other relevant sources. A collaborative team-based approach is adopted to ensure efficient task completion, proper distribution of responsibilities, and effective management of project activities.

2.2 Scrum of Scrums Structure

The project follows a 3-level Scrum of Scrums hierarchy within the broader MODELBD development framework. At Level 1, multiple groups work in parallel on different aspects of Bangladesh's national data sectors and sub-domains. Our group (Group-14) focuses on the Socio-Economic Sector, while other groups work on their respective sectors and sub-sectors. At Level 2, the individual groups are organized into broader sector-specific teams, where related datasets and database models are coordinated and integrated. At Level 3, all sector-level models are integrated into the final MODELBD system. This structure enables parallel development, efficient distribution of responsibilities, independent dataset development, and systematic integration of the different sectoral database models.

Figure **??** illustrates the 3-level Scrum of Scrums hierarchy adopted for the MODELBD project. The bottom layer (Level 1) represents multiple groups working on different sector-specific sub-domains, with our group highlighted as the Socio-Economic Sector team. The middle layer (Level 2) demonstrates the coordination and merging of related groups into broader sector-level teams. The top layer (Level 3) represents the final integration of all sectors into the complete MODELBD system.

2.3 Team Information

Table 2 presents the basic group information, including the group number, group name, and supervisor details.

2.4 Team Structure

The project team follows a collaborative development structure. Responsibilities are distributed among team members according to project requirements, individual expertise, and project contributions. The team members work collaboratively on research, data collection, dataset development, ER modeling, integration, validation, and documentation.

Scrum Master: [**Shamim Hosen (24701035)**]

The Scrum Master is responsible for coordinating project activities, monitoring progress, facilitating communication among team members, assigning tasks, and ensuring that project objectives are completed within the planned timeline.

2.5 Roles and Responsibilities

Table 3 outlines the specific roles and responsibilities of each team member. The Scrum Master coordinates the overall workflow, while other members contribute to research, data collection, dataset development, data transformation, ER modeling, integration, validation, and documentation.

2.6 Product Backlog

The Product Backlog organizes all major project tasks into epics corresponding to the three sprints.

Table 4 presents the Product Backlog organized by epic, showing the sprint assignment, priority level, and description for each major work package.

2.7 Project Timeline

The project is conducted through three development sprints, with each sprint focusing on a distinct phase of the database development process.

Sprint 1 – Research and Data Collection: The first sprint focuses on identifying reliable socio-economic data sources, collecting relevant information, analyzing source materials, and developing structured datasets.

Table 2: Basic Group Information

Group Information	**Details**
Group Number	14
Group Name	UNKNOWN
Supervisor	Prof. Dr. Rudra Pratap Deb Nath

Table 3: Project Roles and Responsibilities

Team Member	**Role**	**Responsibilities**
Shamim Hosen	Scrum Master	Coordinating project activities, monitoring progress, facilitating communication, assigning tasks, and ensuring timeline adherence
Shamim,Forhan, Ashfaq,Usha	Research & Data Collection Team	Identifying socio-economic data sources, collecting data from BBS, BBS HIES, UNDP, UNESCO, World Bank, FAOSTAT, and other relevant sources, and validating collected information
Shamim,Forhan, Usha,Ashfaq	Excel Dataset & Data Transformation Team	Structuring collected information into Excel datasets, cleaning data, standardizing formats, handling missing values, and preparing datasets for database modeling
Shamim,Forhan	ER Diagram & Integration Team	Developing individual ER diagrams, identifying common entities and relationships, reducing redundancy, and integrating individual ER models into the final Socio-Economic ER model
Shamim,Forhan, Ashfaq	Documentation Team	Maintaining project documentation, organizing source references, recording project activities, reviewing outputs, and preparing the final report

Table 4: Product Backlog

Epic	Sprint	Priority	Description
Socio-Economic Data Acquisition	Sprint 1	High	Identify and collect socio-economic data from BBS, BBS HIES, UNDP, UNESCO, World Bank, FAOSTAT, and other reliable sources
Dataset Development	Sprint 1	High	Organize collected socio-economic information into structured Excel datasets and maintain source references
Data Transformation	Sprint 2	High	Clean collected data, standardize naming conventions, handle missing and inconsistent values, and prepare the datasets for database modeling
Entity Identification	Sprint 2	High	Identify major socio-economic entities, attributes, keys, and relationships from the developed datasets
ER Modeling	Sprint 3	High	Develop individual ER diagrams for the selected datasets and validate the models against the collected information
ER Integration	Sprint 3	High	Identify common entities and relationships and integrate the individual ER diagrams into a unified Socio-Economic ER model
Documentation	All Sprints	Medium	Maintain project documentation, source references, sprint activities, diagrams, and prepare the final project report

Sprint 2 – Data Transformation: The second sprint focuses on cleaning and transforming the collected data, standardizing formats and naming conventions, handling missing or inconsistent values, and preparing the datasets for database modeling.

Sprint 3 – ER Modeling and Integration: The third sprint focuses on developing individual ER diagrams, identifying common entities and relationships, integrating the individual ER models, and refining the final Socio-Economic database model.

Figure **??** illustrates the project workflow, showing the development phases from research and source identification to data collection, dataset development, data transformation, ER modeling, integration, and final documentation.

2.8 Tools Used

Throughout the project, we utilize various tools to streamline our workflow, support data processing, and enhance productivity.

Table 5 presents the complete list of tools and technologies utilized throughout the project, including their respective purposes and descriptions.

2.9 Collaboration and Communication

The team uses Google Drive as the primary collaboration platform throughout the project. Google Drive is utilized for sharing datasets, maintaining source materials, storing Excel files, managing ER diagrams, sharing project documentation, and coordinating project updates among team members. Trello is used for project management, including creating task boards, tracking sprint progress, assigning responsibilities, monitoring deadlines, and visualizing workflow. GitHub is used for maintaining project-related files and version history where applicable. These tools enable centralized access to project resources and allow team members to work collaboratively and efficiently.

2.10 Project Management Approach

The project follows a collaborative and iterative development approach based on Scrum methodology. Tasks are distributed among team members according to project requirements, individual expertise, and contributions. Regular communication is maintained to monitor progress, resolve issues, review completed work, and coordinate the integration of datasets and ER models. The Scrum Master coordinates the overall workflow and ensures that milestones

Table 5: Development Tools and Technologies

Tool	**Purpose**	**Description**
Google Drive	Collaboration Platform	Sharing datasets, source materials, ER diagrams, documentation, and project files
Trello	Project Management	Visual task tracking, sprint management, workflow monitoring, and deadline management
Google Sheets / Excel	Dataset Development	Data organization, structuring, cleaning, transformation, and information management
GitHub	Version Control	Maintaining project files, documentation, scripts, and version history
Draw.io	Diagram Design	Creating and visualizing ER diagrams, project figures, and other graphical materials
Overleaf	Documentation	LaTeX-based report generation, formatting, and PDF compilation
ChatGPT	Technical Assistance	Supporting data interpretation, documentation, pattern recognition, coding, and database-related tasks

are achieved according to the planned schedule. Team members continuously review collected data, Excel datasets, ER diagrams, and integration results to improve consistency, completeness, and accuracy. This management approach contributes significantly to the successful development of a comprehensive Socio-Economic database model for Bangladesh.

3 Sprint 1

3.1 Sprint Overview

Sprint 1 was the initial and most important phase of our Socio-Economic Sector database project. The main objective of this sprint was to discover relevant socio-economic data sources, collect raw data, understand the structure and meaning of the collected data, organize the data into structured datasets, and prepare the initial data model for the later stages of the project. The project focuses on socio-economic information of Bangladesh, including population, household characteristics, income and expenditure, employment, poverty, education, health, agriculture, and other related indicators.

The team followed a Scrum-based approach in which the major activities were divided into smaller tasks and tracked through Trello. The primary responsibility for managing and completing the major tasks was taken by Shamim, while Forhan, Usha, and Ashfaq provided support in different activities. Although the work was initially planned to be distributed among all members, Usha unfortunately had to drop out due to an academic year drop, especially near the final stage of the sprint. As a result, a significant portion of the remaining data collection, processing, organization, and documentation work had to be completed by Shamim. Forhan continued to contribute to several data-related tasks, while Ashfaq provided assistance in specific areas when required.

3.2 Sprint Backlog

The Sprint 1 backlog was prepared according to the major requirements of the project. The tasks were tracked through Trello so that progress, responsibility, and completion status could be monitored.

3.3 Data Discovery and Source Identification

The first step of Sprint 1 was to identify reliable sources containing socio-economic information about Bangladesh. Since socio-economic data are distributed among different organizations and statistical platforms, no single source was sufficient to cover all required information.

The major sources considered during the data discovery process were:

- Bangladesh Bureau of Statistics (BBS)
- Bangladesh Bureau of Statistics Household Income and Expenditure Survey (BBS HIES)

Table 6: Sprint 1 Backlog

ID	Task	Priority	Responsible Member(s)	Status
S1-01	Identify reliable socio-economic data sources	High	Shamim, Forhan, Usha, Ashfaq	Done
S1-02	Collect data from BBS and BBS HIES	High	Shamim, Forhan	Done
S1-03	Collect data from UNDP, UNESCO, World Bank and FAOSTAT	High	Shamim, Forhan, Ashfaq	Done
S1-04	Explore additional relevant socio-economic sources	High	Shamim	Done
S1-05	Study the structure and meaning of collected datasets	High	Shamim, Forhan	Done
S1-06	Organize raw data into structured datasets	High	Shamim	Done
S1-07	Clean and standardize collected data	High	Shamim, Forhan	Done
S1-08	Identify entities, attributes and relationships	High	Shamim, Ashfaq	Done
S1-09	Develop initial ER diagrams	High	Shamim	Done
S1-10	Prepare data for population and ETL processing	High	Shamim	Done
S1-11	Organize project files and maintain repository	Medium	Shamim, Forhan	Done
S1-12	Review and document Sprint 1 activities	Medium	Shamim	Done

- United Nations Development Programme (UNDP)
- United Nations Educational, Scientific and Cultural Organization (UNESCO)
- World Bank
- Food and Agriculture Organization Statistical Database (FAOSTAT)
- Other publicly available government, international, and socio-economic data sources

These sources were selected because they provide statistical and socio-economic information covering different dimensions of Bangladesh's population and development. The collected information includes demographic characteristics, household income and expenditure, employment, poverty, education, health, agriculture, food security, and development-related indicators.

3.4 Data Acquisition

After identifying the sources, the next task was to acquire the required datasets. Data were collected from reports, statistical publications, online databases, spreadsheets, PDF documents, and publicly available datasets.

BBS and BBS HIES were particularly important for obtaining Bangladesh-specific household, demographic, income, expenditure, employment, and poverty-related information. International sources such as UNDP, UNESCO, World Bank, and FAOSTAT were used to complement the national datasets with development, education, agriculture, economic, and international indicator data.

The collected files were initially kept in their original form so that the processed datasets could later be compared with the original sources during validation.

3.5 Dataset Organization and Data Population

The raw data collected from different sources were not directly suitable for database modeling because they followed different structures, naming conventions, units, and formats. Therefore, the collected information was organized into structured datasets before being used for further processing.

The datasets were grouped according to their major socio-economic themes, such as:

- Population and demographic information

- Household information
- Income and expenditure
- Employment and labour force
- Poverty and living standards
- Education-related indicators
- Health-related indicators
- Agriculture and food-related statistics
- Economic and development indicators
- Regional and geographical information

During this process, unnecessary columns were identified, column names were made more understandable, missing values were examined, and similar datasets were compared to identify common data elements.

3.6 Initial ER Analysis

After studying the collected datasets, the team started identifying the major entities, attributes, and relationships required for the socio-economic database. The purpose of this stage was to understand how different datasets could be represented within a common database structure.

Common entities and concepts such as population, household, location, trade, food price, displacement, education, and socio-economic indicators were examined. Similar concepts appearing in different datasets were compared carefully to avoid unnecessary duplication. The initial ER analysis also helped identify possible primary keys, foreign keys, relationships, and common attributes that would later be used during the integration and database design stages.

3.6.1 Process of Developing the Final ER Diagram

The final ER diagram was developed gradually from the cleaned and structured Sprint 1 datasets. The main steps followed during the modeling process were:

1. **Identify the main data domains:** The collected datasets were first grouped into geography and census, food price, international trade, and socio-economic and development related information.

2. **Identify candidate entities:** Repeated real-world concepts such as division, district, census record, commodity, exporter, buyer, product, country, displacement event, and education indicator were identified as separate entities.

3. **Identify attributes:** The columns available in the structured datasets were examined and assigned to the most appropriate entities. The attributes were selected so that each entity could store the information required by the source datasets without unnecessary duplication.

4. **Select primary keys:** A unique identifier was selected for each major entity, such as `division_id`, `district_id`, `census_id`, `commodity_id`, and `transaction_id`. These keys uniquely identify individual records.

5. **Identify foreign keys:** Foreign keys were added where one entity depends on another. For example, `division_id` in DISTRICT links a district to its division, while `district_id` in POPULATION_CENSUS, FOOD_PRICE, SDG_INDICATOR, IDP_DATA, DISPLACEMENT_EVENT, and EDUCATION_INDICATOR links those records to a district.

6. **Define relationships and cardinalities:** The relationships between entities were then defined using the primary-key and foreign-key structure. One-to-many relationships were used where one parent record can be associated with multiple child records. Relationship diamonds and the cardinalities `1` and `N` were used in the final Chen-style diagram.

7. **Connect the different domains:** Previously isolated datasets were integrated through common entities. For example, FOOD_PRICE was connected to DISTRICT and COMMODITY, EXPORT_TRANSACTION was connected to EXPORTER, BUYER, DECLARANT, PRODUCT, and COMMODITY, and IMPORT_TRANSACTION was connected to PRODUCT and COUNTRY. This produced a single connected database model instead of several independent diagrams.

8. **Review the complete ER structure:** Finally, the entity names, attributes, key constraints, relationship lines, and cardinalities were checked to ensure that the final diagram represents the collected datasets in a clear and consistent way.

3.6.2 Entities and Attributes in the Final ER Model

The final ER model contains the following major entities and attributes:

1. **DIVISION:** division_id (PK), division_geocode (UNIQUE), division_name.
2. **DISTRICT:** district_id (PK), division_id (FK), district_geocode (UNIQUE), district_name, population_density.
3. **POPULATION_CENSUS:** census_id (PK), district_id (FK), census_year, household_total, population_total, population_male, population_female, population_hijra, population_rural, population_urban, population_slum, and population_floating.
4. **HOUSEHOLD_TYPE:** hh_type_id (PK), census_id (FK), hh_general, hh_institutional, hh_other, pop_general_pct, and pop_institutional_pct.
5. **MARITAL_STATUS:** marital_id (PK), census_id (FK), age_group, never_married, currently_married, widowed, divorced, and separated.
6. **RELIGION_DIST:** religion_id (PK), census_id (FK), muslim_total, hindu_total, christian_total, buddhist_total, and other_religion.
7. **COMMODITY:** commodity_id (PK), commodity_name, category.
8. **FOOD_PRICE:** price_id (PK), district_id (FK), commodity_id (FK), market_name, price_date, price_bdt, unit, and price_type.
9. **EXPORT_TRANSACTION:** transaction_id (PK), exporter_id (FK), buyer_id (FK), product_id (FK), declarant_id (FK), commodity_id (FK), transaction_date, no_of_packages, gross_weight_kg, net_weight_kg, quantity, unit, total_value_usd, total_value_bdt, currency, exchange_rate, delivery_terms, mode_of_transport, month, and year.
10. **EXPORTER:** exporter_id (PK), country_code (FK), exporter_name, address, city, state, and bank_name.
11. **BUYER:** buyer_id (PK), destination_country (FK), buyer_name, and buyer_address.
12. **DECLARANT:** declarant_id (PK), declarant_name, and declarant_address.
13. **PRODUCT:** product_id (PK), hs_code, chapter, heading, sub_heading, description, and package_unit.
14. **COUNTRY:** country_code (PK), country_name, and region.
15. **IMPORT_TRANSACTION:** import_id (PK), product_id (FK), origin_country (FK), import_date, importer_name, quantity, total_value_usd, delivery_terms, and mode_of_transport.

16. **SDG_INDICATOR:** `sdg_id (PK)`, `district_id (FK)`, `indicator_code`, `indicator_name`, `sdg_goal`, `year`, `value`, `unit`, and `area`.
17. **IDP_DATA:** `idp_id (PK)`, `district_id (FK)`, `year`, `stock_displacement`, `flow_displacement`, `conflict_displaced`, and `disaster_displaced`.
18. **DISPLACEMENT_EVENT:** `event_id (PK)`, `district_id (FK)`, `event_name`, `event_type`, `start_date`, `end_date`, `displaced_persons`, and `cause`.
19. **EDUCATION_INDICATOR:** `edu_id (PK)`, `district_id (FK)`, `indicator_name`, `year`, `value`, `sex`, `level`, and `unit`.

3.6.3 Major Relationships and Cardinalities

The final ER diagram uses relationships that follow the identified key structure and the logical connections among the datasets. The major relationships are:

- DIVISION has many DISTRICT records (**1:N**).
- DISTRICT has many POPULATION_CENSUS records (**1:N**).
- POPULATION_CENSUS has detail records in HOUSEHOLD_TYPE, MARITAL_STATUS, and RELIGION_DIST (**1:N**).
- DISTRICT is linked with FOOD_PRICE through `district_id` (**1:N**).
- COMMODITY is linked with FOOD_PRICE through `commodity_id` (**1:N**).
- COMMODITY is linked with EXPORT_TRANSACTION through `commodity_id` (**1:N**).
- EXPORTER, BUYER, DECLARANT, and PRODUCT are linked with EXPORT_TRANSACTION through their respective identifiers (each parent entity to many transaction records, **1:N**).
- COUNTRY is linked with EXPORTER through `country_code` and with BUYER through `destination_country`.
- PRODUCT is linked with IMPORT_TRANSACTION through `product_id` (**1:N**).
- COUNTRY is linked with IMPORT_TRANSACTION through `origin_country` (**1:N**).

- DISTRICT is linked with SDG_INDICATOR, IDP_DATA, DISPLACEMENT_EVENT, and EDUCATION_INDICATOR through `district_id` (**1:N** for each dataset).

These connections ensure that the previously separate datasets are integrated through common geographic, product, commodity, country, and transaction keys. As a result, the final ER model forms one connected structure that can support cross-domain queries and later relational database implementation.

3.6.4 Final ER Diagram

Figure 1 presents the final connected ER diagram prepared from the Sprint 1 datasets.

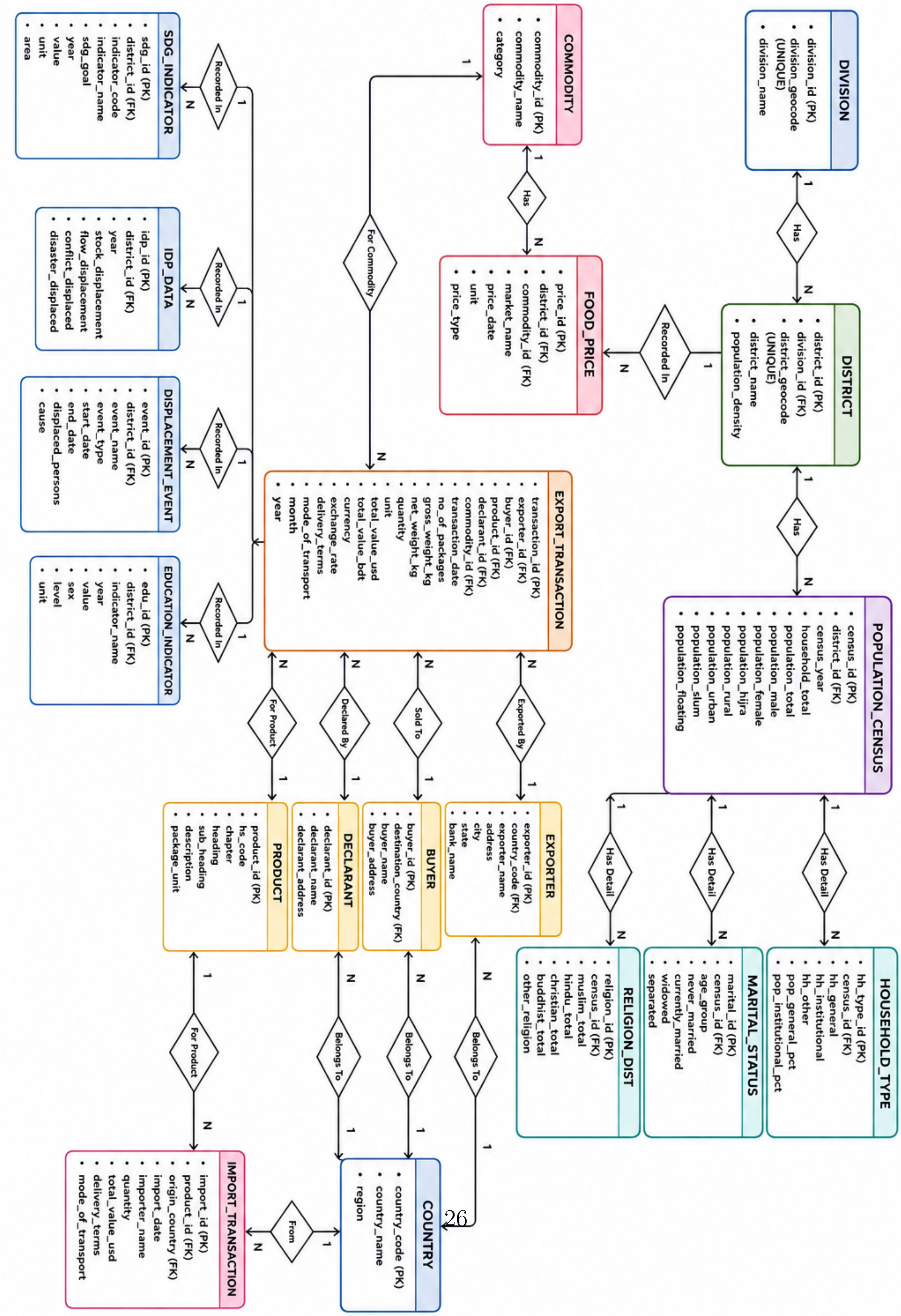

Figure 1: Socio Economics Integrated ER Diagram

3.7 ETL Preparation

An initial ETL workflow was also considered during Sprint 1. The basic process followed was:

1. **Extract:** Collect raw data from BBS, BBS HIES, UNDP, UNESCO, World Bank, FAOSTAT, and other relevant sources.
2. **Transform:** Clean the collected data, standardize names and formats, handle missing values, and organize the datasets into a consistent structure.
3. **Load:** Prepare the transformed datasets for population into the proposed database structure.

The ETL process was developed iteratively because the source datasets were heterogeneous. The extracted and transformed data were retained in organized files so that the process could be repeated and modified when necessary.

3.8 Challenges Faced During Sprint 1

Several challenges were encountered during the completion of Sprint 1. The major challenges are summarized below:

1. **Heterogeneous Data Sources:** Socio-economic data were collected from multiple organizations, and each source followed different formats, definitions, units, naming conventions, geographical classifications, and reporting periods. This made direct integration of the datasets difficult.
2. **Complexity of PDF-Based Data:** A considerable amount of information was available only in PDF reports. Tables were sometimes divided across multiple pages, columns were misaligned, and some values could not be extracted accurately without manual verification.
3. **Missing and Inconsistent Data:** Several datasets contained missing values, duplicate records, inconsistent geographical names, and variations in indicator definitions. These issues required additional checking, cleaning, and standardization before the data could be used.
4. **Data Standardization and Transformation:** Data collected from different sources had to be converted into a consistent structure. Standardizing units, date formats, district names, indicator names, and numerical values required considerable effort.

5. **Source Verification and Reliability:** In some cases, similar indicators had different values across different reports or organizations. Therefore, the original sources, reporting years, definitions, and metadata had to be checked carefully before selecting the appropriate values.

6. **Database and ER Model Design:** Transforming the collected socio-economic datasets into a structured relational database was challenging. Identifying suitable entities, attributes, primary keys, foreign keys, and relationships required repeated analysis to minimize redundancy and maintain data consistency.

7. **Unexpected Change in Team Distribution:** Initially, the tasks were planned to be distributed among all four group members. However, Usha had to discontinue participation because of an academic year drop. This unexpected change significantly affected the original task distribution and increased the workload for the remaining members.

8. **Increased Individual Workload:** Due to the change in team composition, Shamim had to take responsibility for a major portion of the remaining data collection, organization, transformation, ER analysis, and documentation tasks. Forhan continued to assist with several data-related activities, while Ashfaq contributed to specific tasks and discussions whenever required.

9. **Time Management and Final Integration:** The final stage of the sprint involved data validation, database design, diagram preparation, documentation, and formatting within a limited time. Coordinating these activities while maintaining consistency across the report was another significant challenge.

10. **Maintaining Documentation Consistency:** As data were obtained from different sources and processed at different stages, maintaining consistent terminology, table formats, figure references, source citations, and database naming conventions throughout the documentation required additional attention.

3.9 Problem Solving Approach

To address the challenges, the team adopted a combination of automated and manual approaches. Automated techniques were used where possible to extract, reformat, and organize repetitive data, while manually checking important records to ensure accuracy.

Data from multiple sources were compared whenever possible. Standard naming conventions were introduced for common attributes such as divisions, districts, indicators, and categories. Source information was also retained so that important records could be traced back to their original documents during validation.

For ER modeling, similar concepts from different datasets were first identified separately and then compared before deciding whether they should be represented as the same entity or as separate entities.

3.10 Tools and Repository Management

Several tools were used during Sprint 1 to manage the project workflow and data. Trello was used to track tasks, assign responsibilities, and monitor the progress of Sprint 1. Excel and Google Sheets were used to organize and structure the collected datasets.

Python-based data processing techniques were used where necessary for data cleaning, transformation, pattern matching, and repetitive processing tasks. AI-based tools were also used as supporting tools for data extraction, coding assistance, and identifying possible inconsistencies; however, the outputs were manually reviewed before being accepted.

The project files, datasets, processing code, ER diagrams, and documentation were organized in the project's public repository so that the development process and ETL-related resources remain accessible and reproducible.

3.11 Team Contribution

The major responsibility for Sprint 1 was carried by Shamim. Shamim handled most of the source exploration, data collection, dataset organization, cleaning, initial ER analysis, ETL preparation, repository organization, and documentation.

Forhan assisted mainly with data collection, data organization, and several supporting processing tasks. Ashfaq contributed by helping with selected data verification, analysis, and discussions regarding the structure of the database. Usha initially participated in the project activities and contributed during the early stages; however, due to an academic year drop, she was unable to continue until the end of the sprint. As a result, most of the remaining workload had to be completed by Shamim.

Although the workload was not equally distributed in the final stage, the team members contributed according to their availability, and the sprint objectives were completed through continuous communication and cooperation.

3.12 Sprint 1 Outcome

At the end of Sprint 1, the team successfully established the foundation of the Socio-Economic Sector database project. The major outcomes were:

- Identification of major and reliable socio-economic data sources.
- Collection of data from BBS, BBS HIES, UNDP, UNESCO, World Bank, FAOSTAT, and related sources.
- Organization of collected raw data into structured datasets.
- Initial cleaning, standardization, and validation of collected data.
- Identification of major entities, attributes, and relationships.
- Development of initial ER structures based on the collected datasets.
- Preparation of the initial ETL workflow for further data processing.
- Organization of project datasets, source files, codes, and documentation for repository deployment.

3.13 Sprint 1 Retrospective

Sprint 1 provided practical experience in handling real-world socio-economic data rather than idealized textbook datasets. The team learned that data discovery and acquisition require significant time because information is distributed across different sources and often requires manual interpretation.

The sprint also demonstrated the importance of proper task management and communication in a Scrum-based project. The unexpected dropout of one team member created additional workload, particularly for Shamim, but the remaining members continued to provide support where possible. The experience highlighted the importance of maintaining backup task assignments, starting data validation early, and keeping the ETL process and project resources properly organized from the beginning.

4 Deployment and Maintenance

4.1 Overview

Although the primary objective of this project is the development of an integrated conceptual database model rather than a fully implemented software system, proper deployment and maintenance strategies are considered to ensure the long-term usability, accessibility, consistency, and scalability of the developed Socio-Economic Database Model. This chapter discusses the methods used for organizing project artifacts, managing datasets, deploying ETL pipeline code and data in a public repository, maintaining consistency, and preparing the database model for future implementation.

4.2 Public Repository Management

Throughout the project, the collected socio-economic datasets, processed data, ETL pipeline code, source references, Excel files, ER diagrams, integrated ER model, and project documentation are maintained in a public repository. A structured repository hierarchy is used to organize raw data, processed datasets, source files, ETL scripts, ER diagrams, and documentation separately. This approach ensures that all project resources remain accessible, traceable, reusable, and maintainable for future development.

4.3 Dataset Deployment and Organization

The collected socio-economic data are organized into structured datasets according to their respective sources and categories. Data are collected from reliable sources such as Bangladesh Bureau of Statistics (BBS), BBS Household Income and Expenditure Survey (BBS HIES), United Nations Development Programme (UNDP), UNESCO, World Bank, FAOSTAT, and other relevant sources. The processed datasets are maintained alongside the necessary source references to ensure data traceability and facilitate future verification and updates. Where redistribution restrictions apply, only permitted datasets or derived data are included in the public repository.

4.4 ETL Pipeline Deployment

The ETL pipeline is developed to extract, transform, clean, standardize, and prepare socio-economic data collected from multiple sources. All relevant ETL pipeline code is maintained and deployed in the public repository along with the processed datasets and necessary documentation. The extraction process collects data from the identified sources, the transformation process handles inconsistent formats, missing values, duplicate records, and naming variations, and the final processed data are prepared for database modeling and integration. Maintaining the complete ETL workflow in the public repository allows the data processing process to be reviewed, reproduced, maintained, and extended in future.

4.5 Maintenance of Socio-Economic Data

Socio-economic data may change over time due to the publication of new surveys, reports, statistical updates, and revised indicators. Therefore, the collected datasets are periodically reviewed and updated based on the latest available information from the respective sources. New datasets can be incorporated into the existing repository structure, while outdated or inconsistent information can be identified and corrected through the ETL pipeline. Source references and metadata are also maintained to preserve the origin and context of the collected data.

4.6 Maintenance of ER Models

During the project lifecycle, individual ER diagrams and the integrated Socio-Economic ER model undergo several revisions and refinements. The maintenance process involves entity validation to ensure that entities accurately represent the underlying socio-economic data, attribute verification to reduce redundancy and maintain consistency, relationship refinement to incorporate newly identified relationships, and integration updates whenever changes occur in individual ER diagrams. This iterative maintenance process improves the consistency and reliability of the final integrated conceptual model.

4.7 Data Consistency and Quality Assurance

Maintaining data quality is a critical component of the project. Several measures are taken to ensure consistency, including verification of collected information from reliable sources, cross-checking related information from multiple sources, identification and removal of duplicate records, standardization of naming conventions and data formats, handling of missing or inconsistent values, and validation of entities and relationships before integration. These practices help reduce data anomalies and improve the accuracy and reliability of the final Socio-Economic database model.

4.8 Future Database Deployment

Although the project primarily focuses on conceptual database design and data integration, the final integrated ER model can be implemented in a relational database management system in the future. Possible implementation platforms include MySQL, PostgreSQL, Oracle Database, and Microsoft SQL Server. The integrated ER model can be transformed into relational schemas, normalized tables, primary and foreign key constraints, and SQL-based database structures. The resulting database can support socio-economic data management, statistical analysis, development planning, research, policy analysis, and decision-support applications.

4.9 Scalability Considerations

The developed database model is designed to support the integration of additional socio-economic datasets, indicators, organizations, geographic regions, and time periods. Future enhancements may include more detailed population and demographic information, household income and expenditure, employment and labor statistics, poverty indicators, agricultural statistics, health and education indicators, and regional development data. The modular structure of the integrated ER model and the maintained ETL pipeline allow new datasets and data sources to be incorporated without major changes to the existing structure.

4.10 Summary

The deployment and maintenance strategies adopted in this project ensure the effective organization, accessibility, reproducibility, and long-term management of socio-economic data and database artifacts. By deploying the ETL pipeline code and permitted datasets in a public repository, the project provides a transparent and reusable foundation for future development. Although a fully implemented production database is beyond the current scope, the developed conceptual model, datasets, and ETL workflow provide a strong foundation for future deployment, expansion, and data-driven analysis of Bangladesh's socio-economic information.

5 Collaboration

5.1 Overview

Collaboration played a crucial role in the successful completion of this project. Since the project involved extensive research, data collection, dataset development, data transformation, ETL pipeline development, ER diagram creation, and integration of socio-economic information from multiple sources, effective teamwork was essential for managing the workload and maintaining data quality.

The project was completed through a collaborative team-based approach where responsibilities were distributed among team members according to project requirements and individual contributions. In addition to collaboration within our own group, coordination with other groups and different levels of the Scrum of Scrums structure was important for ensuring consistency and supporting the integration of the Socio-Economic Sector into the broader MODELBD framework.

5.2 Team Communication

Regular communication was maintained among team members throughout the project lifecycle. Communication activities included:

- Discussing project requirements and objectives
- Planning research and data collection activities
- Identifying reliable socio-economic data sources
- Reviewing collected data and dataset structures
- Discussing data cleaning and transformation strategies
- Reviewing ETL pipeline outputs
- Discussing entity and relationship identification
- Reviewing and refining ER diagram designs
- Resolving data inconsistency and integration issues
- Monitoring project progress and sprint activities

Regular communication helped ensure that all members remained informed about project developments and contributed effectively to the project objectives.

5.3 Intra-Level Collaboration

Intra-level collaboration refers to the communication and coordination among members of Group 14 working on the Socio-Economic Sector. Team members were assigned different responsibilities such as research and data collection, dataset development, data transformation, ETL processing, ER modeling, integration, and documentation.

Members regularly shared collected information, discussed data inconsistencies, reviewed each other's work, and provided feedback before finalizing datasets and ER models. This internal collaboration helped distribute the workload and reduced the possibility of errors in the final database model.

5.4 Inter-Level Collaboration

Inter-level collaboration involved communication and coordination with other groups and broader levels of the Scrum of Scrums structure. Since Group 14's Socio-Economic Sector is intended to be integrated into the larger MOD-ELBD framework, coordination with other sector-specific groups was important for identifying common entities, attributes, relationships, and data requirements.

Discussions with other groups helped us understand how socio-economic information could relate to other national sectors such as education, health, agriculture, and other development-related domains. This collaboration also helped identify possible overlapping entities and avoid unnecessary duplication during the integration process.

5.5 Collaboration Platform

Google Drive served as one of the primary collaboration platforms for the project. The platform was used for:

- Sharing research materials
- Storing source references
- Managing Excel and Google Sheets datasets
- Sharing processed datasets
- Sharing ETL-related resources
- Maintaining ER diagrams
- Maintaining project documents

- Coordinating team contributions

Trello was used for task management, sprint planning, progress tracking, and responsibility assignment. GitHub was used where applicable for maintaining project files, ETL pipeline code, documentation, and version history. These platforms enabled centralized access to project resources and improved coordination among team members.

5.6 Positive Collaboration Experiences

Several positive experiences were observed during the collaboration process.

5.6.1 Effective Task Distribution

Project tasks were distributed among team members according to their responsibilities and areas of contribution. This reduced workload concentration and allowed different parts of the project to progress in parallel.

5.6.2 Shared Knowledge

Team members regularly exchanged knowledge regarding socio-economic data sources, data interpretation, data transformation, ETL processing, database modeling, and ER diagram design. This knowledge sharing improved the overall quality of the project and helped members understand areas outside their individual responsibilities.

5.6.3 Cooperative Problem Solving

Several challenges were encountered while working with data from BBS, BBS HIES, UNDP, UNESCO, World Bank, FAOSTAT, and other sources. Problems related to inconsistent formats, missing values, duplicate information, source interpretation, and entity identification were addressed through group discussions and collective decision-making.

5.6.4 Inter-Group Learning

Interaction with other groups provided opportunities to learn how different sectors were being modeled and integrated. Comparing approaches helped us identify common database concepts and understand how the Socio-Economic Sector could interact with other sectors within the MODELBD framework.

5.6.5 Timely Completion of Tasks

Continuous communication and coordination helped the team complete major project activities within the planned schedule. Regular progress monitoring also helped identify unfinished tasks and redistribute responsibilities when necessary.

5.7 Challenges Faced During Collaboration

Like most collaborative projects, several challenges were encountered during the development process.

5.7.1 Large and Heterogeneous Data Sources

Socio-economic information was collected from multiple sources, and the structure, terminology, units, and formats were not always consistent. Coordinating the processing and validation of such heterogeneous data required significant effort and communication.

5.7.2 Data Consistency Issues

Information from different sources sometimes contained different naming conventions, time periods, units, or values for similar indicators. Team members needed to discuss and verify these differences before including the information in the final datasets.

5.7.3 ETL and Data Transformation Complexity

Cleaning and transforming data from different sources required careful coordination. Some datasets required additional processing, manual verification, or modification of the ETL workflow before they could be used for database modeling.

5.7.4 ER Integration Complexity

Integrating individual ER diagrams into a comprehensive Socio-Economic model required careful analysis of common entities and relationships. Different team members sometimes proposed different interpretations of the same data structure, which required additional discussion and review.

5.7.5 Coordination with Other Groups

Inter-group communication was sometimes challenging because different groups followed different schedules, priorities, and modeling approaches. Maintaining consistency between sector-specific models required additional coordination.

Despite these challenges, the team successfully addressed most issues through communication, cooperation, repeated review, and collective decision-making.

5.8 Conflict Resolution

Minor disagreements occasionally occurred regarding entity definitions, attribute selection, relationship structures, data interpretation, ETL processing strategies, and integration approaches.

These issues were resolved through:

- Open discussion among team members
- Reviewing the original data sources
- Comparing alternative modeling approaches
- Testing different data transformation strategies
- Evaluating the requirements of the integrated model
- Reaching consensus through group decisions

The Scrum Master facilitated discussions when necessary and helped maintain project progress. Decisions were generally made by considering data evidence, database design principles, project requirements, and the future integration needs of the MODELBD framework.

5.9 The Role of AI in Our Journey

Throughout the project, AI tools were used as supporting resources for tasks such as data interpretation, pattern identification, text processing, ETL assistance, documentation, and technical problem solving. AI tools helped reduce the time required for repetitive tasks and provided suggestions for handling complex data-processing and database-related problems.

However, AI was not treated as a replacement for human judgment. Several limitations were observed during the project:

- AI sometimes missed information during PDF and document extraction.
- AI occasionally misunderstood the context of socio-economic indicators.
- AI-generated data-processing code sometimes required manual modification.
- AI could produce technically valid results that were not always appropriate for the specific dataset.
- Source verification and final decisions still required human judgment.

AI therefore acted as an assisting tool rather than the primary decision-maker. The team remained responsible for verifying information, understanding the context of the data, validating the results, and making final database design decisions.

5.10 Lessons Learned from Collaboration

The project provided valuable experience in collaborative database development and project management. The key lessons learned include:

1. Importance of clear and regular communication
2. Benefits of structured task distribution
3. Importance of validating data before integration
4. Necessity of maintaining proper source references
5. Importance of version control and shared repositories
6. Value of teamwork in solving complex data-processing problems
7. Importance of coordination between different Scrum levels
8. Necessity of reviewing ER models before final integration
9. Importance of human judgment when using AI-assisted tools

These experiences improved both the technical and interpersonal skills of the team members and provided practical knowledge about collaborative database development.

5.11 Summary

The collaborative approach adopted by Group 14 contributed significantly to the successful development of the Socio-Economic Database Model. Effective intra-level communication helped the team manage research, data collection, ETL processing, ER modeling, and documentation, while inter-level collaboration supported coordination with other groups and the broader MODELBD framework.

Although the project involved challenges such as heterogeneous data sources, data inconsistencies, ETL complexity, ER integration, and inter-group coordination, these challenges provided valuable learning experiences. The project demonstrated the importance of teamwork, communication, data validation, shared resources, and coordinated decision-making in developing an integrated database model for Bangladesh's socio-economic information.

6 Comments on Course Pedagogy

6.1 Overview

The Database System (CSE 413) course provides both theoretical and practical knowledge about database design, data modeling, data integration, and database management. The course pedagogy combines theoretical concepts with a practical project, allowing students to apply database theories and methodologies to real-world data problems. The project-based approach provided an opportunity to work with heterogeneous socio-economic data collected from sources such as BBS, BBS HIES, UNDP, UNESCO, World Bank, FAOSTAT, and other relevant sources. The structured project workflow, Scrum-based collaboration, and defined deliverables created an environment that was closer to practical software and data engineering activities than traditional classroom exercises.

6.2 Opportunities and Limitations

The project-based pedagogy provided several opportunities that are difficult to achieve through conventional lecture-based courses. Working with real-world socio-economic data exposed us to data inconsistency, missing values, heterogeneous formats, source verification, ETL processing, ER modeling, and data integration. The Scrum of Scrums structure also provided an opportunity to experience collaboration among multiple teams working toward a common large-scale system.

However, the pedagogy also has some limitations. The limited project duration can make it difficult to perform extensive data validation and refinement. Working with heterogeneous real-world sources is significantly more time-consuming than working with prepared academic datasets. Furthermore, coordination between different groups may sometimes create communication and integration challenges.

Table 7 summarizes the major opportunities and limitations experienced during the project.

6.3 Comparison with Other Course Pedagogies

Compared with courses that primarily depend on lectures, examinations, and predefined problem sets, the Database System project provided a more practical and open-ended learning environment. In previous courses, datasets or problems are often well-defined, whereas this project required us to identify sources, understand raw information, clean the data, determine appropriate entities and relationships, and develop a conceptual model based on real-world requirements.

Table 8 compares the project-based pedagogy of the Database System course with more traditional course approaches.

6.4 Lessons Learned

The project provided several important lessons beyond the technical concepts of database systems. Working with real socio-economic data demonstrated that practical database development involves much more than creating tables and relationships. Data understanding, source verification, communication, task management, conflict resolution, documentation, and continuous validation are equally important.

Table 9 summarizes the major lessons learned from the project in technical, managerial, and collaborative areas.

6.5 Industry Experience and Collaboration

The project provided an industry-like experience by requiring us to work with real-world data, multiple stakeholders, distributed responsibilities, deadlines, and integration requirements. The Scrum of Scrums structure introduced the concept of working within a larger development ecosystem where individual teams are responsible for specific components but must consider the requirements of the overall system.

Working on the Socio-Economic Sector also demonstrated that data engineering and database development require continuous communication between data collectors, data processors, model designers, and integration teams. This experience helped us understand the importance of coordination, responsibility, accountability, and communication in professional software and data-related projects.

6.6 Suggestions for Future Batches

Based on our experience, several improvements could make the project more effective for future batches. Providing a clear integration guideline before the beginning of the project would help different groups follow consistent naming conventions, entity definitions, and relationship standards. A small sample dataset and ETL demonstration before Sprint 1 could also help students understand the expected workflow.

Table 10 presents the major suggestions for improving the project-based pedagogy for future batches.

6.7 Summary

The Database System course and its associated project provided valuable learning experiences in both technical and non-technical areas. The Socio-Economic Sector project strengthened our understanding of database design, ER modeling, data integration, ETL processing, data validation, teamwork, project management, conflict resolution, and technical documentation. Compared with conventional academic exercises, the project provided greater exposure to real-world data and industry-like collaboration. With improvements such as clearer integration guidelines, additional ETL training, better inter-group coordination, and more time for data validation, the pedagogy can provide an even more effective learning experience for future batches.

Table 7: Opportunities and Limitations of the Course Pedagogy

Opportunities	**Limitations**
Working with real-world socio-economic datasets instead of only theoretical examples	Limited project duration restricts extensive data validation and refinement
Experience with data collected from multiple national and international sources	Data extraction and cleaning from heterogeneous sources require significant time
Practical experience with ETL, data transformation, and database integration	Inconsistent formats and missing values increase the complexity of data processing
Scrum of Scrums provides experience of large-scale team collaboration	Coordination between multiple groups can sometimes be difficult
Development of individual and integrated ER models provides practical database modeling experience	Integration requirements between groups may not always be clear at the beginning
Use of project management and collaboration tools creates an industry-like working environment	Learning multiple tools simultaneously can increase the initial workload

Table 8: Comparison of Course Pedagogy

Aspect	**Traditional Course Approach**	**Database System Project Approach**
Data	Mostly predefined and structured	Real-world and heterogeneous data from multiple sources
Problem Definition	Usually clearly provided by the instructor	Requires research and interpretation of real-world requirements
Teamwork	Often limited to small group activities	Continuous collaboration within and across Scrum groups
Data Processing	Limited practical data cleaning	Extensive data collection, cleaning, transformation, and validation
Project Management	Less emphasis on formal management	Uses Scrum, sprint planning, task distribution, and progress monitoring
Integration	Usually focuses on individual solutions	Requires integration of multiple datasets, ER models, and sector-level outputs
Industry Exposure	Mostly theoretical or laboratory-based	Provides an industry-like experience through real-world data and team collaboration

Table 9: Major Lessons Learned from the Project

Area	**Lesson Learned**
Industry Vibe	Real-world data are often incomplete, inconsistent, and distributed across multiple sources, requiring continuous investigation and validation.
Collaboration	Large projects require clear task distribution, regular communication, shared resources, and coordination among multiple teams.
Conflict Resolution	Technical disagreements can be resolved through discussion, source verification, comparison of alternatives, and consensus-based decisions.
Project Management	Sprint-based planning and clear task priorities help manage large and complex project requirements efficiently.
Data Management	Data cleaning, transformation, standardization, and validation are essential before developing a reliable database model.
Documentation	Proper documentation and source tracking are important for maintaining reproducibility and future maintenance.
AI-Assisted Development	AI tools can accelerate repetitive and technical tasks, but human verification and domain understanding remain essential.

Table 10: Suggestions for Improving Course Pedagogy

Area	**Suggestion**
Integration Guideline	Provide a common integration contract or database modeling guideline before the first sprint begins so that different groups can follow a consistent structure.
Transition from Prerequisite Coursework	The prerequisite coursework on disease and museum datasets provided useful foundational experience. A short bridging session could further demonstrate how the same data-processing concepts can be applied to larger, heterogeneous, real-world socio-economic datasets.
Real-World Data Complexity	Include a brief discussion on handling practical issues such as inconsistent formats, missing values, conflicting definitions, geographical naming differences, and data collected from multiple organizations.
Data Collection Time	Allocate sufficient time for data collection, cleaning, standardization, and validation because real-world datasets often require considerable manual verification.
Inter-Group Coordination	Arrange periodic coordination sessions among different Scrum levels to identify and resolve data integration and schema compatibility issues at an earlier stage.
Version Control	Provide or reinforce basic Git/GitHub workflow practices so that groups can systematically maintain ETL scripts, database files, documentation, and dataset versions.
Evaluation	Evaluate both the final output and the development process, including data quality, documentation, collaboration, database design, and project management.

7 Conclusion and Future Work

7.1 Conclusion

This project successfully develops an integrated conceptual database model for the Socio-Economic Sector of Bangladesh. The primary problem addressed by the project is the fragmentation of socio-economic information across multiple organizations, reports, surveys, websites, and datasets, which makes data collection, management, comparison, and analysis difficult. To address this problem, the project involves extensive research, data collection, dataset development, data cleaning and transformation, ETL pipeline development, Entity-Relationship (ER) modeling, and integration of multiple socio-economic datasets.

The project uses data collected from reliable national and international sources including Bangladesh Bureau of Statistics (BBS), BBS Household Income and Expenditure Survey (BBS HIES), United Nations Development Programme (UNDP), UNESCO, World Bank, FAOSTAT, and other relevant sources. The collected information is organized into structured datasets and processed to handle inconsistent formats, missing values, duplicate information, and different naming conventions. Individual ER diagrams are then developed based on the identified datasets and subsequently integrated into a comprehensive Socio-Economic Sector ER model.

The final integrated model provides a structured conceptual representation of socio-economic information and establishes relationships among different socio-economic entities, indicators, organizations, geographic areas, and statistical information. The project demonstrates how heterogeneous real-world data can be systematically collected, transformed, organized, and modeled using database concepts.

The Reality: Working with real-world socio-economic data was significantly more challenging than working with predefined academic datasets. Data were often distributed across different sources, formats were inconsistent, and extraction and validation required considerable effort. However, these challenges provided valuable practical experience and demonstrated that real-world database development requires not only technical knowledge but also data understanding, critical thinking, validation, collaboration, and continuous decision-making.

7.2 Project Outcomes

The major outcomes of the project include:

- Identification and collection of socio-economic data from reliable national and international sources.
- Development and organization of structured datasets from sources such as BBS, BBS HIES, UNDP, UNESCO, World Bank, and FAOSTAT.
- Development of an ETL workflow for extracting, cleaning, transforming, and preparing heterogeneous socio-economic data.
- Identification of common entities, attributes, and relationships across multiple datasets.
- Development of individual Entity-Relationship diagrams for different socio-economic datasets.
- Integration of individual ER models into a comprehensive Socio-Economic Sector ER model.
- Improvement of practical skills in data collection, data transformation, database modeling, and data integration.
- Development of teamwork, collaboration, project management, and conflict-resolution skills.
- Deployment of relevant datasets, ETL pipeline code, and project artifacts in a public repository for accessibility and future development.

These outcomes demonstrate that the major objectives of the project have been successfully addressed and provide a foundation for future implementation of a comprehensive socio-economic database system.

7.3 Significance of the Project

The significance of this project lies in its ability to provide a unified conceptual structure for socio-economic information that is currently distributed across multiple sources. An integrated database model can make it easier to organize, retrieve, compare, and analyze socio-economic information from different domains and time periods.

The developed model can potentially support researchers, students, policy makers, development organizations, and other stakeholders in understanding socio-economic trends and relationships. It also demonstrates the practical application of database theories, conceptual modeling, data integration, and ETL techniques to a real-world national-level problem.

7.4 Limitations

Although the project achieves its primary objectives, several limitations remain. The project is primarily focused on conceptual database design and data integration rather than the development of a fully functional production database system. Therefore, the physical implementation of the complete database, application interface, and real-time data services are beyond the current scope.

The availability and quality of socio-economic data also vary across different sources. Some datasets contain missing values, inconsistent formats, different measurement units, or different reporting periods, which can affect the completeness and comparability of the integrated data. Furthermore, socio-economic indicators and statistics are continuously updated, meaning that the developed datasets require periodic maintenance and revision.

Another limitation is that the current model may not cover every possible socio-economic indicator or organization in Bangladesh. Due to project time constraints, some potentially relevant datasets and sources could not be incorporated. In addition, further normalization, physical schema optimization, performance testing, and implementation-level validation remain necessary before deploying the model as a production database.

7.5 Reflections on AI Assistance

Throughout the project, AI tools were used as supporting resources for data processing, pattern identification, ETL assistance, documentation, and technical problem solving. AI helped reduce the time required for repetitive tasks and provided useful suggestions for handling large amounts of information.

However, AI was treated as a supporting tool rather than a replacement for human judgment. AI-generated outputs sometimes contained extraction errors, misunderstood the context of socio-economic indicators, or required modification to work correctly with specific datasets. Therefore, the team had to manually verify sources, validate processed data, review generated code, and make final decisions based on the actual requirements and context of the project.

This experience demonstrated that AI can significantly amplify productivity, but reliable database development still requires human understanding, critical thinking, source verification, and quality assurance.

7.6 Future Work

Several opportunities exist for extending the project in the future.

7.6.1 Database Implementation

The conceptual ER model can be transformed into a fully functional relational database using database management systems such as MySQL, PostgreSQL, Oracle Database, or Microsoft SQL Server. The implementation can include normalized relational schemas, primary and foreign key constraints, indexes, views, triggers, and appropriate integrity constraints.

7.6.2 Normalization and Optimization

The current conceptual model can be further analyzed and normalized to reduce redundancy and prevent insertion, deletion, and update anomalies. Query optimization, indexing, and database performance testing can also be performed after physical implementation.

7.6.3 Web-Based Socio-Economic Information System

A web-based information system can be developed using the database as its backend. Such a system could allow users to search, filter, compare, and visualize socio-economic information based on indicators, geographic regions, organizations, and time periods.

7.6.4 Expansion of Data Coverage

Future development can incorporate additional socio-economic domains and datasets, including more detailed population and demographic statistics, employment and labor information, poverty and income indicators, health and education statistics, agricultural information, regional development indicators, and other relevant national statistics.

7.6.5 Automated ETL Pipeline

The existing ETL workflow can be further developed into an automated pipeline that periodically retrieves updated information from supported sources, performs data validation and transformation, and updates the database automatically. This would reduce manual data collection and improve the timeliness of the system.

7.6.6 Data Visualization and Analytics

The database can be integrated with data visualization and business intelligence tools to provide dashboards, statistical reports, trend analysis, and comparative analysis. Such analytical capabilities could help researchers and policy makers understand socio-economic changes and support evidence-based decision-making.

7.6.7 Integration with Government and International Data Systems

Future work may include integration with additional government databases and international statistical systems. Such integration could provide a more comprehensive and continuously updated representation of Bangladesh's socio-economic condition.

7.7 Final Remarks

The Socio-Economic Database project provides valuable practical experience in database design, data integration, ETL processing, data validation, teamwork, and project management. By applying the theoretical concepts learned in the Database System (CSE 413) course to real-world socio-economic data, the project demonstrates the importance of structured data management and conceptual modeling in addressing complex information-management problems.

The final integrated Socio-Economic ER model, processed datasets, and ETL workflow provide a strong foundation for future database implementation, analytical applications, and data-driven decision-making. The knowledge and experience gained through this project will be valuable for future academic, research, and professional activities in database systems, data engineering, and software development.

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